MILITARY STANDARD

MIL-STD-1320-177A
(NAVY)
27 MARCH 1991

TRUCKLOADING

5.125 CARTRIDGE MK 182, MK 186, MK 193,
MK 214, MK 216, MK 229 OR MK 232
IN PALLET ADAPTER MK 144
UNIT LOAD FOR UNDERWAY REPLENISHMENT

SUPERSEDING
MIL-STD-1320-177
3 APRIL 1979

UNIT LOAD DATA

UNIT LOAD DOCUMENT MIL-STD-1323-201
-298, -313 AND -370

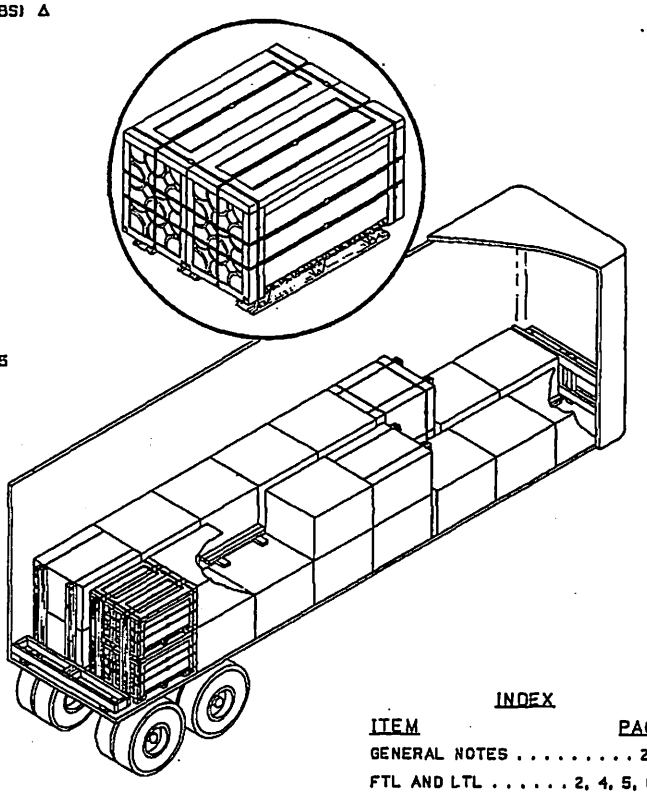
DIMENSIONS . . . . 53 3/8 L X 42 1/4 W X 32 1/4 H

CUBE . . . . . 42.1 CU FT

Δ DO NOT USE FOR SHIPPING WEIGHT

CARTRIDGE MK-MOD	DODIC/ NALC	UNIT LOAD MIL-STD-1323	UNIT LOAD WEIGHT (LBS) Δ
182-1	EW76	-298	1662
182-2	7W69	-298	1516
186-0	EW75	-201	1614
186-1	EW75	-201	1614
193 (ALL UP)	EW77	-298	1660
193 (BODY)	GW95	-298	1612
193 (PROP)	GW96	-298	1420
214-0 OR 1	9W22	-313	1662
216-0	3W80	-313	1790
229-0	4W61	-370	1658
232-0	5W95	-370	1658

FOR HAZARD CLASSIFICATIONS, SEE NAVSEA OP 2165



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL AND LTL	2, 4, 5, 6 AND 7
LOAD TABLES	3 AND 4
UNITIZING DETAIL	7

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE:
DISTRIBUTION IS UNLIMITED.

AMSC NO. - N/A

REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
			APPROVAL	
A	ADDED MK'S 182-1, 182-2, 186-1, 193-0 214-0, 214-1, 216-0, 229-0 AND 232-0. DELETED MK182-0. REVISED PAGE 1, GENERAL NOTES AND TABLE IV.	3/27/91	[illegible]	[illegible]

NOTES:
1. UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED
FOR HIGHWAY SHIPMENT ONLY.

S/ M. S. GRAY NWHC 4/2/79
SIGNATURE TDA, WPNSTA EARLE DATE

S/ J. E. KELLY 4/2/79
SIGNATURE SEASYSCOM BY DIRECTION DATE

ORIGINATOR S/ S. R. PETOIA 3/30/79
DATE

WPNSTA EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:46Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING UNIT LOADS FOR UNDERWAY REPLENISHMENT OF 5.125 CARTRIDGES IN MK 144 ADAPTERS. SEE TABLE ON PAGE 1 FOR LISTING OF CARTRIDGES AND UNIT LOADS.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40, 42, 44 AND 45 FT VAN TRAILERS WITH BOTH NAILABLE AND NON-NAILABLE FLOORS. TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER).
3. LENGTH OF THE STRUTS, PIECE 5 IN DETAIL A SHALL BE SUFFICIENT TO CAUSE THE BLOCKING ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE TRAILER DOORS WHEN THE DOORS ARE CLOSED.

WARNING

DO NOT USE TRAILERS WITH ROLL UP DOORS.

4. TO PREVENT LONGITUDINAL MOVEMENT OF THE UNIT LOADS IN THE SECOND LAYER, THE UNIT LOADS SHALL BE UNITIZED (SEE UNITIZING DETAIL, PAGE 7) WHERE THE PATTERN CHANGES FROM ONE LAYER HIGH TO TWO LAYERS HIGH OR AT THE REAR OF THE LOAD. STIFFENERS SHALL FACE THE SINGLE LAYER SIDE OR THE REAR WHEN AT THE TRAILER DOOR. WHENEVER THERE IS A SINGLE STACKED UNIT LOAD (SEE TABLE I FOR 17 UNIT LOADS) THE STIFFENERS SHALL BE ON BOTH ENDS OF THE UNIT LOAD.
5. WHEN THE TRAILER HAS A NON-NAILABLE FLOOR, SUBSTITUTE SWAY BRACE DETAIL C OR E, FOR SLEEPERS AND INSTALL BETWEEN BOTTOM LAYER PALLETS AS PRESCRIBED IN NOTE 6 BELOW.
6. THE SWAY BRACE, DETAIL C OR E, IS ASSEMBLED IN PLACE. INSERT SUPPORTS, PIECE 7, AGAINST INBOARD SIDE OF PALLETS VERTICAL SUPPORTS, POSITION STRINGERS, PIECE 8, AGAINST PALLET VERTICAL SUPPORT AND NAIL IN PLACE WITH THREE 10d NAILS. WHEN USING SWAY BRACE IN LIEU OF SLEEPERS, PIECE 9, INSTALL BETWEEN LOWER LAYER PALLETS IN SAME MANNER.
7. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA, CLOSE AND SEAL TRAILER DOORS AND ATTACH APPLICABLE "EXPLOSIVES" PLACARDS TO BOTH SIDES, FRONT AND BACK OF TRAILER.
9. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I, HEAVY DUTY, FINISH A, B OR C.
10. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL AND LTL 40, 42, 44 AND 45 FT TRAILERS (VAN)

WARNING

THIS STANDARD PROVIDES FOR INCREASED LOADINGS WHICH PRODUCE HEAVIER GROSS TANDEM AXLE WEIGHTS AND GROSS VEHICLE WEIGHTS. THESE INCREASED WEIGHTS ARE NOT AUTHORIZED IN ALL 50 STATES. BEFORE LOADING MORE THAN 26 UNIT LOADS IN A TRAILER, DETERMINE THAT THE LIMITATIONS OF THE NOTES IN TABLES I, II AND III ARE MET.

1. TABLES I, II AND III PROVIDE THE CORRECT LOAD PATTERNS FOR LENGTH OF THE TRAILER AND THE TARE WEIGHT OF THE TRACTOR-TRAILER BEING LOADED.

NOTE

LIGHT WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEM AXLES 6,800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 10,500 LBS MAXIMUM.
STANDARD WEIGHT TRACTOR-TRAILER - TARE WEIGHT OF TRAILER TANDEMS AXLES 7,800 LBS MAXIMUM AND THE DRIVE TANDEM AXLES 11,500 LBS MAXIMUM.

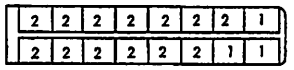
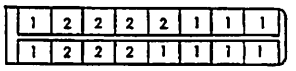
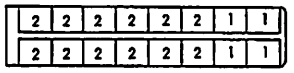
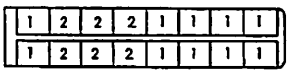
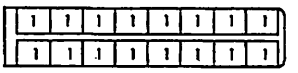
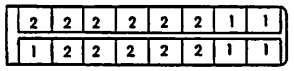
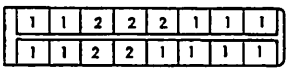
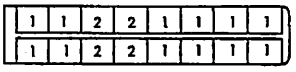
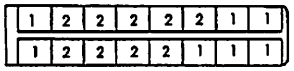
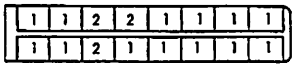
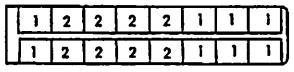
THE ABOVE MAXIMUM WEIGHTS MAY BE ADJUSTED WHEN EXPERIENCE INDICATES THAT THE WEIGHT RESTRICTIONS CAN BE MET.

2. TO DETERMINE THE CORRECT LOAD PATTERN TO USE, SELECT THE CORRECT TABLE FOR THE LENGTH OF TRAILER BEING LOADED. OPPOSITE THE NUMBER OF UNIT LOADS TO BE LOADED, FIND THE CORRECT LOAD PATTERN. MAKE CERTAIN THAT THE QUANTITY BEING SHIPPED IS COMPATIBLE WITH THE TARE WEIGHT OF THE TRACTOR-TRAILER BEING LOADED AND THE WEIGHT LAWS OF THE STATES THROUGH WHICH THE SHIPMENT IS ROUTED. BLOCK AND BRACE USING THE PRINCIPLES SHOWN ON PAGES 5, 6 AND 7.
3. THE TRUCKLOADS SHOWN ON PAGES 5, 6 AND 7 DO NOT LIMIT SHIPMENTS TO THE NUMBER OF UNIT LOADS SHOWN. THEY ARE PRESENTED TO SHOW THE BASIC PRINCIPLES OF BLOCKING ONLY. THE LOAD PATTERN TABLES SHOW THE CORRECT NUMBER OF UNIT LOADS FOR THE LENGTH AND WEIGHT OF THE TRAILER BEING LOADED AND THE WEIGHT RESTRICTIONS OF THE ROUTING.
4. WHEN LOADING TRAILERS WITH A SECOND LAYER, FOLLOW THE PROCEDURES ON PAGES 5 AND 6 DECREASING THE NUMBER OF UNIT LOADS IN THE SECOND LAYER AS REQUIRED.
5. WHEN SHIPPING LESS THAN THE AMOUNT SHOWN IN THE LOAD TABLES, FOLLOW THE PROCEDURES SHOWN ON PAGE 7. DO NOT USE TRAILER WITH NON-NAILABLE FLOORS.
6. WHEN A SECOND LAYER UNIT LOAD DOES NOT HAVE A UNIT LOAD OPPOSITE IT, USE LATERAL STRAPS AS SHOWN ON PAGES 5 AND 6.

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:46Z
Check the source to verify that this is the current version before use.

TABLE I

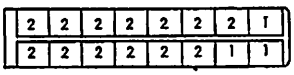
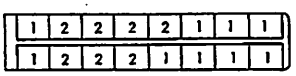
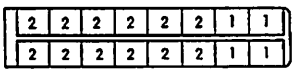
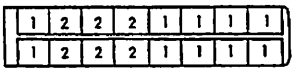
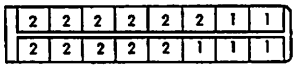
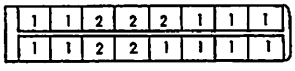
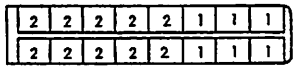
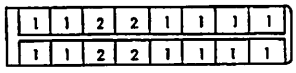
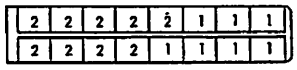
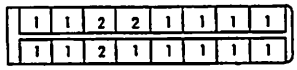
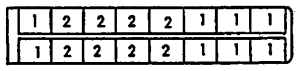
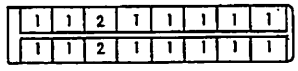
40 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN FWD →	NO. OF UNIT LOADS	LOAD PATTERN FWD →	NO. OF UNIT LOADS	LOAD PATTERN FWD →
*** 29		23		17	
** 28		22		16	
* 27		21		15	
26		20		14	
25		19		13 OR LESS USE DOUBLE ROW PATTERN	
24		18			

NOTE: NUMBERS IN BLOCKS INDICATE NUMBER OF UNIT LOADS IN A LAYER

TABLE II

42 FT TRAILER

NO. OF UNIT LOADS	LOAD PATTERN FWD →	NO. OF UNIT LOADS	LOAD PATTERN FWD →	NO. OF UNIT LOADS	LOAD PATTERN FWD →
*** 29		23		17	
** 28		22		16	
* 27		21		15	
26		20		14	
25		19		13 OR LESS USE DOUBLE ROW PATTERN	
24		18			

NOTE: NUMBERS IN BLOCKS INDICATE NUMBER OF UNIT LOADS IN LAYER.

*** AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. ONLY LIGHT WEIGHT TRACTOR-TRAILERS SHALL BE USED.

** AUTHORIZED WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS OR MORE ON THE TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80,000 LBS. BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS MAY BE USED.

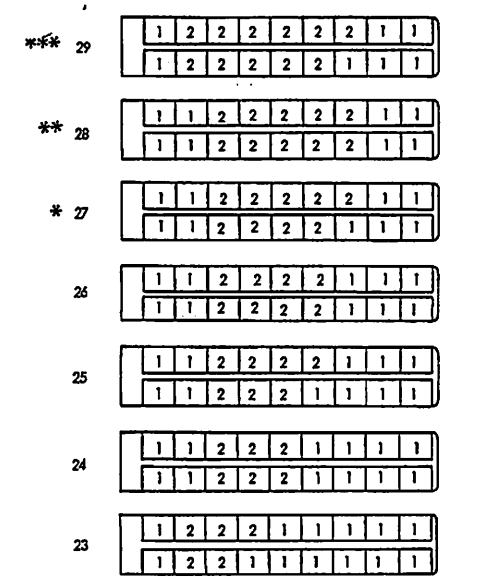
* AUTHORIZED FOR LIGHT WEIGHT TRACTOR-TRAILERS ONLY WHEN ROUTING IS THROUGH STATES LIMITING TANDEM AXLES TO 32,000 LBS AND GROSS VEHICLE WEIGHTS TO 73,280 LBS. AUTHORIZED FOR BOTH STANDARD AND LIGHT WEIGHT TRACTOR-TRAILERS WHEN ROUTING IS THROUGH STATES PERMITTING 34,000 LBS ON TANDEM AXLES AND GROSS VEHICLE WEIGHTS OF 80,000 LBS.

NOTE: WHEN SHIPPING 26 OR LESS UNIT LOADS, BOTH STANDARD AND LIGHTWEIGHT TRACTOR-TRAILERS MAY BE USED IN ALL 50 STATES WITHOUT EXCEEDING THE WEIGHT LAWS.

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:46Z
Check the source to verify that this is the current version before use.

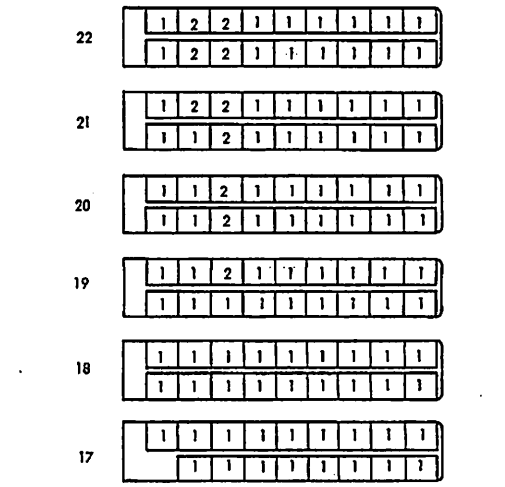
TABLE III
44 & 45 FT TRAILERS

NO. OF UNIT LOADS — LOAD PATTERN — FWD →



44 & 45 FT TRAILERS

NO. OF UNIT LOADS — LOAD PATTERN — FWD →

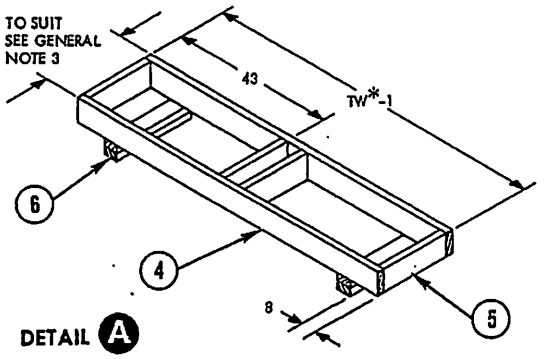


16 OR LESS: USE DOUBLE ROW PATTERN

*** ** * SEE PAGE 3

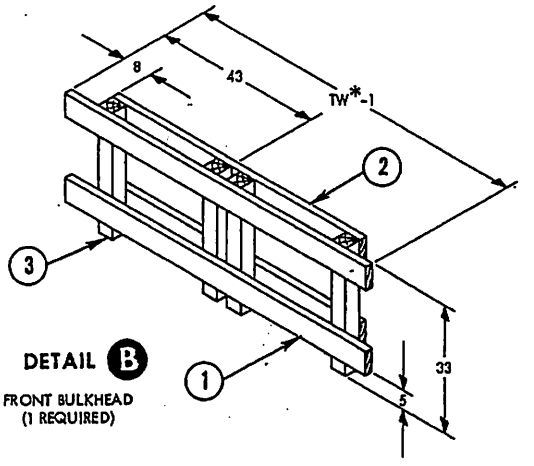
NOTE: NUMBERS IN BLOCKS INDICATE NUMBER OF UNIT LOADS IN A LAYER.

*TW = TRAILER WIDTH



DETAIL A

REAR BLOCKING ASSEMBLY (1 REQUIRED)



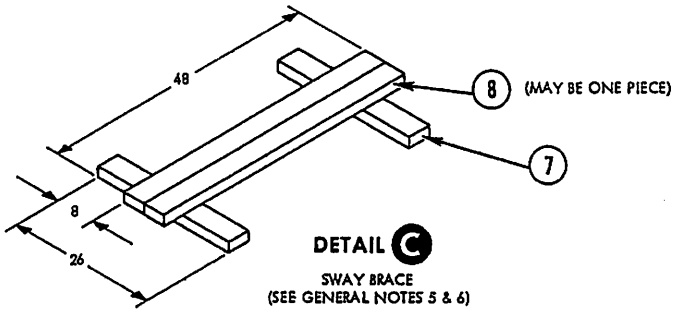
DETAIL B

FRONT BULKHEAD (1 REQUIRED)

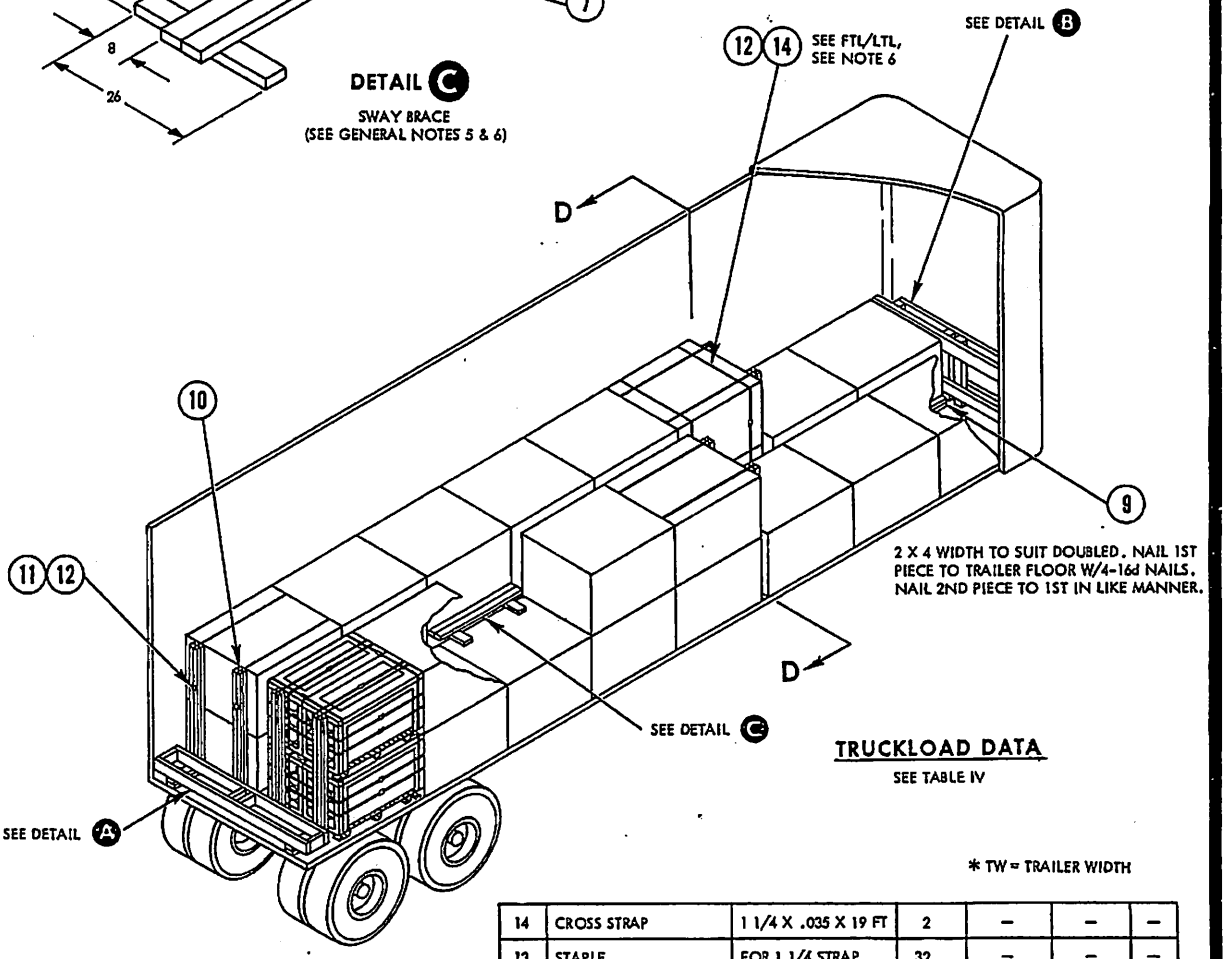
TABLE IV
TRUCKLOAD DATA

NUMBER OF UNIT LOADS	TRUCKLOAD WEIGHT (LBS) APPROXIMATE									
	MK 182-1	MK 182-2	MK 186-0 OR MK186-1	MK 193 ALL UP CTDG	MK 193 BODY ASS'Y	MK 193 PROP. ASS'Y	MK 214-0 OR MK 214-1	MK 216-0	MK 229-0	MK 232-0
29	48,198	43,964	46,806	48,140	46,748	41,180	48,198	------	48,082	48,082
28	46,536	42,448	45,192	46,480	45,136	39,760	46,536	------	46,424	46,424
27	44,874	40,932	43,578	44,820	43,524	38,340	44,874	48,330	44,766	44,766
26	43,212	39,416	41,964	43,160	41,912	36,920	43,212	46,540	43,108	43,108
25	41,550	37,900	40,350	41,500	40,300	35,500	41,550	44,750	41,450	41,450
24	39,888	36,384	38,736	39,840	38,688	34,080	39,888	42,960	39,792	39,792
23	38,226	34,868	37,122	38,180	37,076	32,660	38,226	41,170	38,134	38,134

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:46Z
Check the source to verify that this is the current version before use.

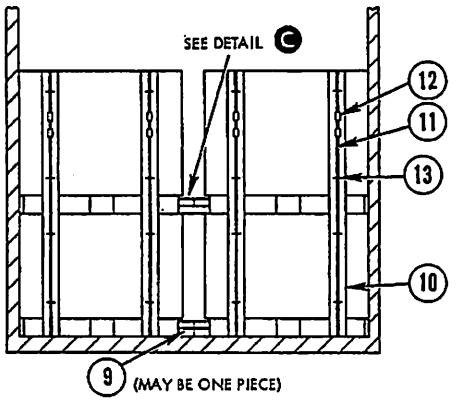


DETAIL C
SWAY BRACE
(SEE GENERAL NOTES 5 & 6)



TRUCKLOAD DATA
SEE TABLE IV

* TW = TRAILER WIDTH

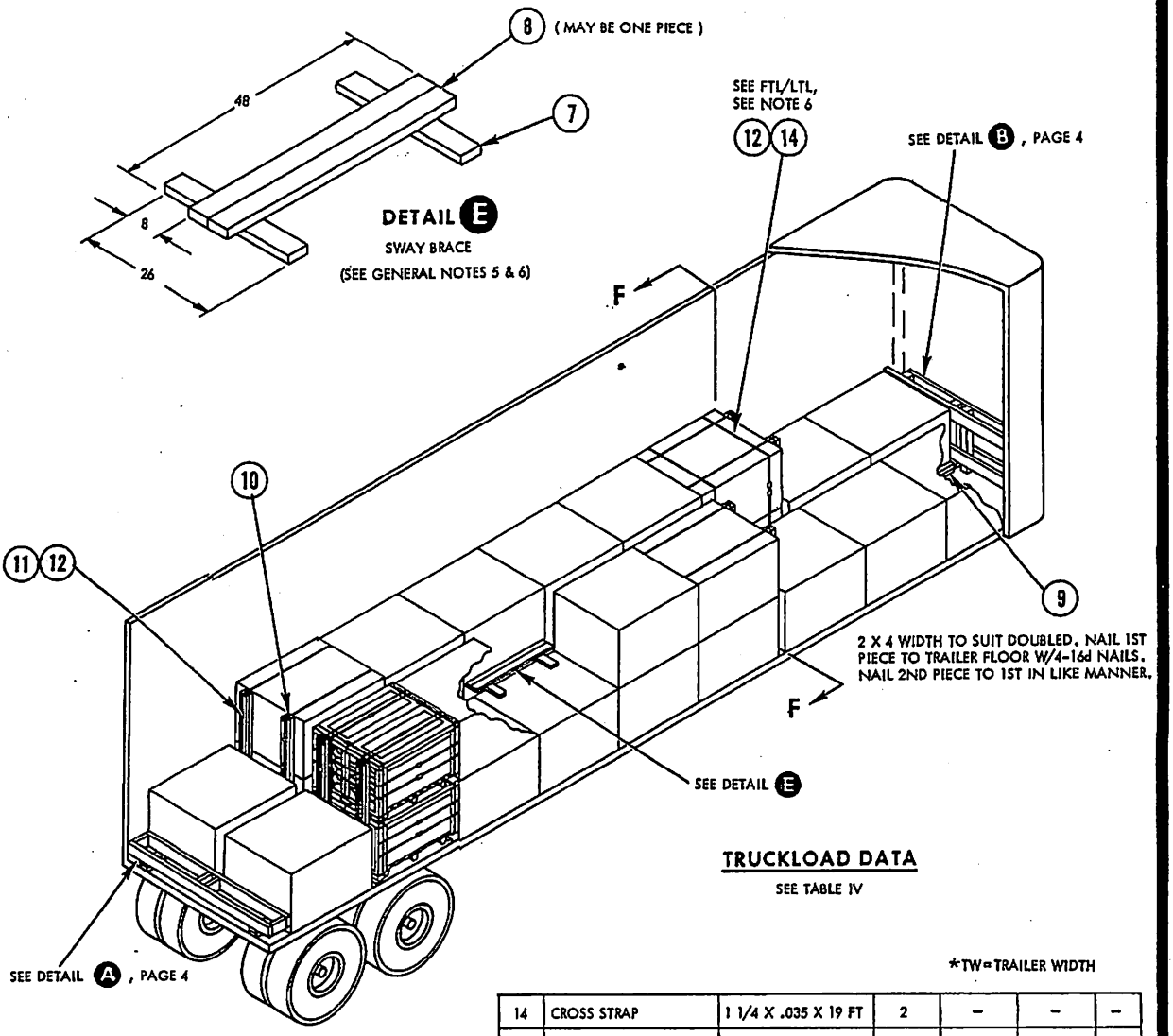


SECTION D-D

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
14	CROSS STRAP	1 1/4 X .035 X 19 FT	2	–	–	–
13	STAPLE	FOR 1 1/4 STRAP	32	–	–	–
12	SEAL	FOR 1 1/4 STRAP	20	–	–	–
11	STRAP	1 1/4 X .035 X 21 FT	8	–	–	–
10	STIFFENER	2 X 6 X 59	16	DBL LAMINATE	4	10d
9	SLEEPER	2 X WIDTH TO SUIT X 48	AS REQ'D	SEE FIELD NOTE		
8	STRINGER	2 X WIDTH TO SUIT X 48	10	7	3/JOINT	10d
7	SUPPORT	2 X 4 X 24	10	SEE 8	–	–
6	SUPPORT	2 X 4 TO SUIT	4	4	2/JOINT	10d
				6	3/JOINT	10d
5	STRUT	2 X 6 TO SUIT	4	SEE 4	–	–
4	CROSSMEMBER	2 X 6 X TW*-1	2	5	3/JOINT	16d
3	VERTICAL	4 X 4 X 33	4	SEE 1 & 2	–	–
2	FWD CROSSMEMBER	2 X 6 TO SUIT	2	3	3/JOINT	12d
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	3	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:46Z
Check the source to verify that this is the current version before use.



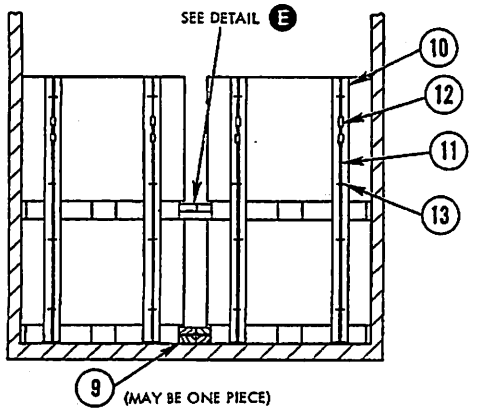
TRUCKLOAD DATA

SEE TABLE IV

*TW=TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
14	CROSS STRAP	1 1/4 X .035 X 19 FT	2	–	–	–
13	STAPLE	FOR 1 1/4 STRAP	32	–	–	–
12	SEAL	FOR 1 1/4 STRAP	20	–	–	–
11	STRAP	1 1/4 X .035 X 21 FT	8	–	–	–
10	STIFFENER	2 X 6 X 59	16	DBL LAMINATE	4	10d
9	SLEEPER	2 X WIDTH TO SUIT X 48	AS REQ'D	SEE FIELD NOTE		
8	STRINGER	2 X WIDTH TO SUIT X 48	10	7	3/JOINT	10d
7	SUPPORT	2 X 4 X 24	10	SEE 8	–	–
6	SUPPORT	2 X 4 TO SUIT	4	4 6	2/JOINT 3/JOINT	10d 10d
5	STRUT	2 X 6 TO SUIT	4	SEE 4	–	–
4	CROSSMEMBER	2 X 6 X TW*-1	2	5	3/JOINT	16d
3	VERTICAL	4 X 4 X 33	4	SEE 1 & 2	–	–
2	FWD CROSSMEMBER	2 X 6 TO SUIT	2	3	3/JOINT	12d
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	3	3/JOINT	12d

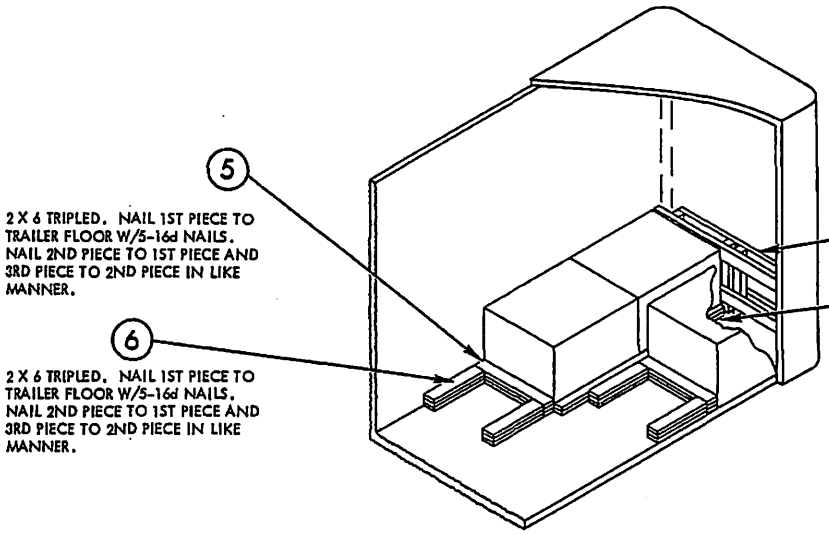
LIST OF MATERIALS & NAILING DATA



SECTION F-F

29 UNIT LOAD ON 45 FT TRAILER

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:46Z
Check the source to verify that this is the current version before use.

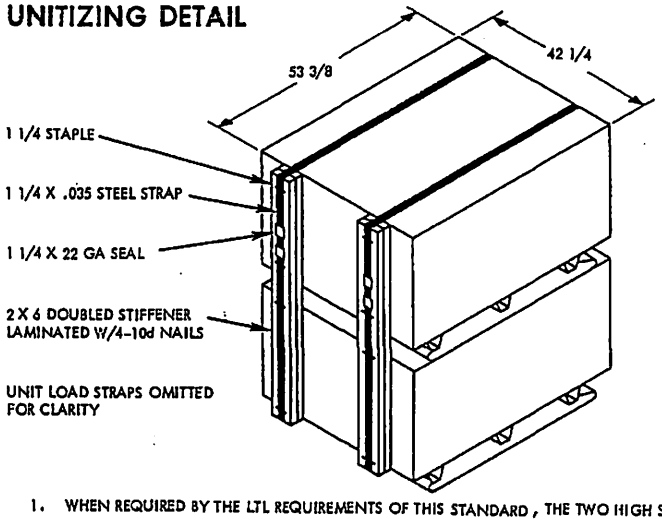


*TW = TRAILER WIDTH

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
6	BACKUP CLEAT	2 X 6 X 30	12	SEE FIELD NOTE		
5	CROSSMEMBER	2 X 6 X 42	6	SEE FIELD NOTE		
4	SLEEPER	2 X WIDTH TO SUIT X 48	AS REQ'D	SEE FIELD NOTE		
3	VERTICAL	4 X 4 X 33	4	SEE 1 & 2	–	–
2	FWD CROSSMEMBER	2 X 6 TO SUIT	2	3	3/JOINT	12d
1	AFT CROSSMEMBER	2 X 6 X TW*-1	2	3	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

UNITIZING DETAIL



NOTE

WHEN LOADING A TRAILER WITH A SINGLE STACK OF TWO HIGH UNIT LOADS, THE STIFFENERS SHALL BE AT BOTH ENDS OF THE UNITIZED LOADS. SEE GENERAL NOTE 4.

1. WHEN REQUIRED BY THE LTL REQUIREMENTS OF THIS STANDARD, THE TWO HIGH STACKED UNIT LOADS SHALL BE STRAPPED AS SHOWN ABOVE.
2. THE 2 X 6 DOUBLED STIFFENER SHALL EXTEND TO THE TOP OF STACKED UNIT LOADS AND TO THE PALLET OF THE BOTTOM UNIT LOAD. THE STIFFENER SHALL BE LAMINATED WITH FOUR 10d NAILS TO PREVENT SHIFTING.
3. THE 1 1/4 INCH STEEL STRAPS POSITION AS SHOWN ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE DECK OF THE BOTTOM PALLET WITH THE STRAPS HOLDING THE STIFFENERS IN PLACE.
4. TENSION STRAPS AND SEAL WITH TWO DOUBLE CRIMPED SEALS OR ONE DOUBLE NOTCHED SEAL. SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STAPLES.

REVIEW ACTIVITY:
NAVY-OS

PREPARING ACTIVITY:
NAVY-OS
(PROJECT NO. 8140-N693)

Source: https://assist.dla.mil -- Downloaded: 202610-23T23:46Z
Check the source to verify that this is the current version before use.